EXPERIMENT - 10

Application Performance Optimization

Library Book Issue Management

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Subject	Low-Code Application Development using Zoho Creator
Application	Library Book Issue Management
Platform	Zoho Creator
Record Date	23 October 2026

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Aim

To apply query tuning and bounded subform input to the library project, and evaluate indexing considerations while preserving the required request form.

Experiment overview

Write a one-line overview of what you compared and tested in your own application.

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1. Problem Statement

A queue count does not need every request to be loaded and inspected in Deluge. Fetching all rows becomes unnecessary work as the library grows. A staff screen also needs only a short, ordered list of recent requests.

2. Tools and Data

Item	Value
Application / environment	
Source form	
Test function	
Dataset	
Test method	

3. Implementation Procedure

1. *Describe the step you performed.*
2. *Describe the step you performed.*
3. *Describe the step you performed.*
4. *Describe the step you performed.*
5. *Describe the step you performed.*

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4. Deluge Implementation

Paste your own function code here (baseline loop, criteria-based count, sorted and bounded fetch).

5. Test Results

Check	Observed result
Baseline rows iterated	
Baseline / optimized Pending count	
Counts match	
Bounded rows returned / limit	
Writes performed	

State what these results do and do not prove (correctness vs. measured performance).

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6. Indexing Review

Summarise what you found about how Creator handles indexing, and whether you created any index.

Candidate	Decision for this library app
Record ID	
Request_Status	
Preferred_Issue_Date	
Register_Roll_Number	

7. Bounded Subform Implementation

Describe the form, subform fields and the maximum row limit you set.

Test	Actual result
Test 1 (single book line)	
Test 2 (several book lines)	
Attempt to exceed the row limit	
Persistence in report	

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Implementation Evidence

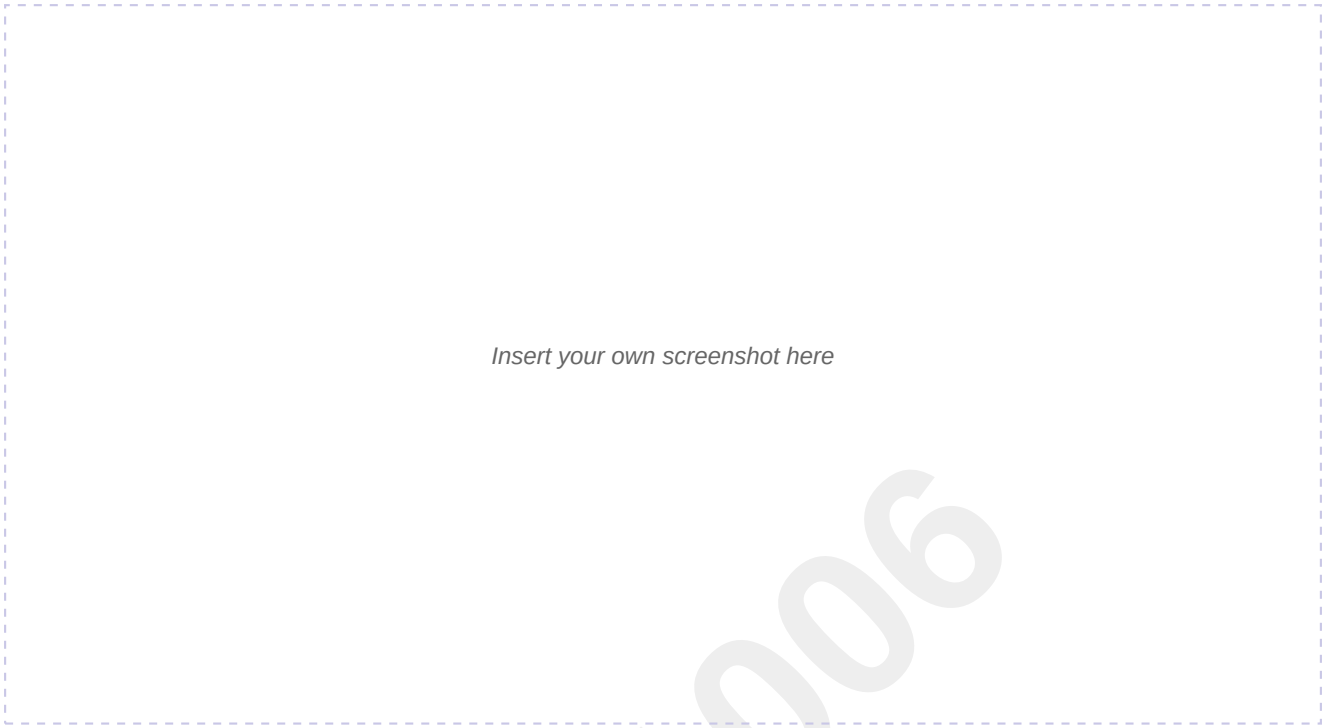

Figure 10.1. Development function execution output.

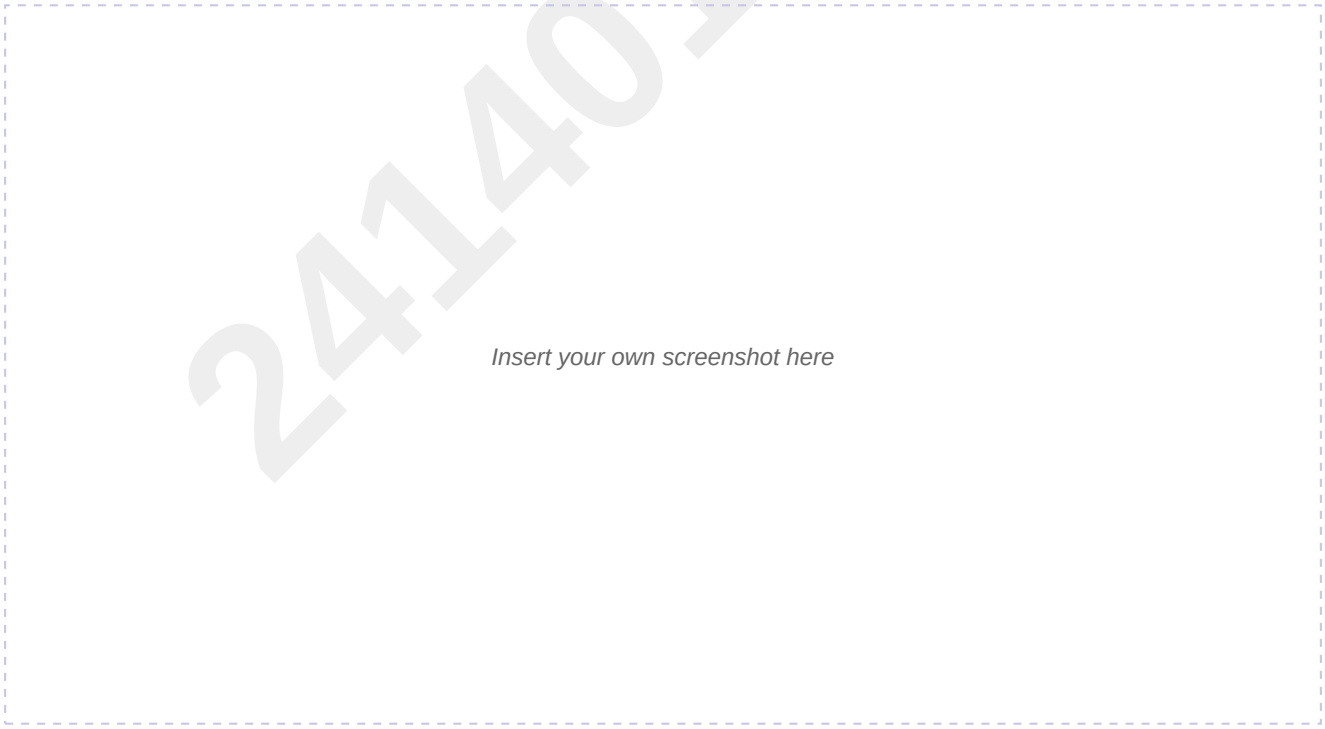

Figure 10.2. Development subform with mandatory child fields and row cap.

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Implementation Evidence

Insert your own screenshot here

Figure 10.3. Negative test: attempt to exceed the row limit.

Insert your own screenshot here

Figure 10.4. Saved parent and child records in the report.

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8. Result

Summarise what you implemented and what your tests showed.

9. Learning Outcomes

1. Use criteria-based aggregate functions for counts and totals.
2. Limit and sort records needed by a screen.
3. Separate correctness evidence from performance benchmarking.
4. Choose subforms only when the business relationship requires child records.

10. Further Performance Test

For a production-scale assessment, use approved synthetic data, compare median and slow-percentile response times under the same conditions, and inspect the slowest workflows. Record dataset size, concurrency, environment and API calls. Do not infer large-scale performance from a handful of records.

References

1. Zoho Creator platform performance
2. Deluge: fetching and limiting records

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